YUSO Thermodynamics

2018

Testing Packet

Competitor Names:

School:

Team Number:

Part 1. (3 pts each - 15 pts total)

1. Calculate the point where the temperature given in Fahrenheit is the same as the temperature given in Celsius.

2. The record low temperature for New Haven is -5℉ for February and -19℃ for December. Which one is lower?

3. What is the mathematical relation between the Boltzmann's Constant (B) and the Gas Constant (K).

4. Express the gas constant in SI base units. (given R=0.0082atm*L/mol*K and 1 atm = 101,325 Pa)

5. Which of the following is the standard unit for heat according to the SI system?
   A. Joule
   B. Small Calorie
   C. Big Calorie
   D. British Thermal Unit

Part 2 (40 pts total)

1. One experiment conducted by British physicist Joseph Black demonstrated the existence of latent heat. Black mixed equal parts ice at 0°C with water at 78°C and obtained water at 0°C. Calculate the latent heat of fusion for water, expressed in kJ/kg. (The specific heat for water is 1 cal/g) (7 pts)

2. We obtain two cylindrical thermal conductors. Conductor A has a radius of 10 cm and a length of 50 cm, while Conductor B has a radius of 5 cm and a length of 25 cm. What is the relationship between the thermal conductances of the two conductors K_A and K_B? (3 pts)
   A. $K_A = 4K_B$
   B. $K_A = 2K_B$
   C. $K_A = K_B$
   D. $2K_A = K_B$
   E. $4K_A = K_B$

3. Gallium's melting point is only 29.8°C, its molar mass is 69.72 U, and its latent heat for the solid-liquid phase transition is 5.59 kJ/mol. The specific heat of solid Gallium is 0.37 J/(gm*K). Using blackbody radiation, we estimate that a typical human hand outputs 0.39 watts of thermodynamic power. If you were to grasp 1 mol of solid Gallium at 20°C at this temperature, how long would it take to completely melt in your hand? (7 pts)

4. In this problem, we explore one of the most famous paradoxes in thermodynamics - the entropy of mixing.

   For this problem, you will need the Sackur-Tetrode equation for an ideal gas:

   $$\Delta S = nR \ln (V_f / V_i)$$

   where $R = 8.31$

   a) Consider a box with two identical chambers separated with removable partition. We add 2 moles each of two distinguishable (different) ideal gases to each side, then remove the partition. Recalling that entropy is an extensive variable, what is the total change in entropy of this system? (10 pts)

b) (This question will be graded for completion and cogency, not accuracy.) Now consider the same situation, but each chamber is filled with 2 moles of the same ideal gas (indistinguishable). Is the total change of entropy the same? Should this be the case? (10 pts)

This is a problem that has driven physicists mad for years, and is one of the earliest examples where classical physics failed. This paradox was not resolved until quantum and statistical mechanics came to be.

5. For an ideal gas conforming to Maxwell–Boltzmann distribution, rank its mean speed, root mean square speed and most probable speed in ascending order. (3 pts)

Part 3 (50 pts total)

Imagine an engine comprised of an ideal gas with four states. Each of these states has a set of thermodynamic properties: temperature, pressure, and volume (T,P,V). The engine follows these steps between its states:

$$1 \rightarrow 2, \text{ isothermal expansion } [\ T_1 = T_2\ ,\ P_1V_1 = P_2V_2\ ]$$

$$2 \rightarrow 3, \text{ adiabatic expansion } [\ \Delta Q = 0\ ,\ P_2V_2^{3/2} = P_3V_3^{3/2}\ ]$$

$$3 \rightarrow 4, \text{ isothermal compression } [\ T_3 = T_4,\ P_3V_3 = P_4V_4\ ]$$

$$4 \rightarrow 1, \text{ adiabatic compression } [\ \Delta Q = 0\ ,\ P_4V_4^{3/2} = P_1V_1^{3/2}\ ]$$

- The system consists of 1 mol of ideal gas.
- The heat capacity of the ideal gas is 12.5 J / (mol·K)
- The hot temperature for isothermal expansion is 432 K, while the cold temperature for isothermal compression is 400 K.

1. Draw a P-V diagram for this system. Be sure to label each state, step, and axis clearly in your diagram. Note that this diagram does not need to be to scale. (10 pts)

2. Calculate the total internal energy change for both adiabatic processes (indicate which stages you are calculating) (10 pts)

3. How does the internal energy of the system change after one cycle? Answer quantitatively, if possible, and provide a description. (10 pts)

4. Is heat added to or removed from the system in Step 1 (1 → 2)? In Step 3 (3 → 4)? (10 pts)

5. What is the total efficiency of the system? (10 pts)

Part 4 (3 pts each)

1. During this man's time at Yale, he coined the word "statistical mechanics" and published his seminal work, "On the Equilibrium of Heterogeneous Substances," which established the concept of a chemical potential and the foundations of enthalpy. Due to his contributions to understanding phase transitions and entropy, he is also the namesake of a thermodynamic free energy. Albert Einstein called him "the greatest mind in American history." Who is he?

2. This engineer developed the first complete theory of the steam engine, and laid the foundation for all heat engines. He succeeded in finding the relationship between saturated vapour pressure and temperature, and lated used this to relate temperature, pressure and density of gases, as well as to derive expressions for the latent heat of evaporation of a liquid. He is also a namesake of a temperature scale. Who is he?

3. This physicist and mathematician is considered one of the founders of thermodynamics. In his most important paper, "On the Moving Force of Heat", he published in basic ideas of the second law of thermodynamics, and later used this to introduce the concept of entropy. Along with another man, he is the namesake of a famous equation used to estimate the vapor pressure at any temperature, an extremely handy equation to use for phase transitions. Who is he?

4. Circle one: Joule discovered the universality of the conversion between electrical and thermal energy.

   True False

5. Circle one: Who first proposed the Second Law of Thermodynamics?
   A. Clausius

   B. Joule

   C. Ampere

   D. Maxwell

   E. Carnot